

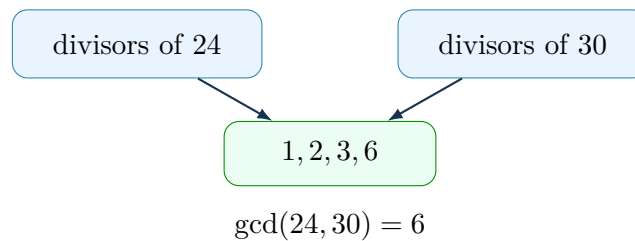
Common Divisors and GCD

1 Core Definitions

Common Divisor

An integer d is a **common divisor** of a and b when

$$d \mid a \quad \text{and} \quad d \mid b.$$



Always true

The integer 1 is a common divisor of any two integers.

2 Linear Combination Rule

Common Divisor Closure

If $d \mid a$, $d \mid b$, and $x, y \in \mathbb{Z}$, then

$$d \mid (ax + by).$$

Two useful special cases are

$$d \mid (a + b) \quad \text{and} \quad d \mid (a - b).$$

Example

Since $6 \mid 24$ and $6 \mid 30$,

$$6 \mid (30 - 24) = 6,$$

and

$$6 \mid (2 \cdot 24 - 30) = 18.$$

3 Greatest Common Divisor

Greatest Common Divisor

For integers a and b , not both zero, the **greatest common divisor** is the largest positive common divisor of a and b . It is written

$$\gcd(a, b).$$

Example	Meaning
$\gcd(24, 30) = 6$	The common divisors are 1, 2, 3, 6; the largest is 6.
$\gcd(5, 7) = 1$	The only positive common divisor is 1.
$\gcd(0, 9) = 9$	Every integer divides 0, so the largest positive divisor of 9 is 9.
$\gcd(0, 0) = 0$	Defined this way so gcd formulas work uniformly.

Range

If $a \neq 0$ and $b \neq 0$, then

$$1 \leq \gcd(a, b) \leq \min(|a|, |b|).$$

4 Basic GCD Properties

Property	Meaning
$\gcd(a, b) = \gcd(b, a)$	Order does not matter.
$\gcd(a, b) = \gcd(-a, b)$	Sign of one input does not matter.
$\gcd(a, b) = \gcd(a , b)$	GCD is controlled by absolute values.
$\gcd(a, 0) = a $	The nonzero number is the answer.
$\gcd(a, ka) = a $	A number and any integer multiple of it have gcd $ a $.

5 Bézout Characterization

Theorem

If a and b are integers, not both zero, then $\gcd(a, b)$ is the smallest positive integer of the form

$$ax + by, \quad x, y \in \mathbb{Z}.$$

The expression $ax + by$ is called an **integer linear combination** of a and b .

Proof Setup

Let

$$S = \{ax + by : x, y \in \mathbb{Z}, ax + by > 0\}.$$

Since a and b are not both zero, S contains at least one positive integer. Let

$$s = \min S.$$

Then for some $x_0, y_0 \in \mathbb{Z}$,

$$s = ax_0 + by_0.$$

Step 1: Show that $s \mid a$

Divide a by s . By the division theorem, there exist integers q and r such that

$$a = qs + r, \quad 0 \leq r < s.$$

So

$$r = a - qs.$$

Substitute $s = ax_0 + by_0$:

$$r = a - q(ax_0 + by_0).$$

Distribute:

$$r = a - qax_0 - qby_0.$$

Group the a -term and the b -term:

$$r = a(1 - qx_0) + b(-qy_0).$$

Thus r is also an integer linear combination of a and b .

But $0 \leq r < s$, and s is the smallest positive linear combination. Therefore r cannot be positive. Hence

$$r = 0.$$

So

$$a = qs,$$

which means

$$s \mid a.$$

Step 2: Show that $s \mid b$

The same argument with b gives

$$b = qs + r, \quad 0 \leq r < s.$$

Using $s = ax_0 + by_0$,

$$r = b - q(ax_0 + by_0) = a(-qx_0) + b(1 - qy_0).$$

Again, r is a linear combination of a and b . Since $0 \leq r < s$, the minimality of s forces

$$r = 0.$$

Therefore,

$$\boxed{s \mid b.}$$

Conclusion so far

Since $s \mid a$ and $s \mid b$, the number s is a common divisor of a and b . Therefore,

$$\boxed{s \leq \gcd(a, b).}$$

Step 3: Show that $\gcd(a, b) \leq s$

Let

$$g = \gcd(a, b).$$

By definition,

$$g \mid a \quad \text{and} \quad g \mid b.$$

So g divides every integer linear combination:

$$g \mid (ax + by) \quad (x, y \in \mathbb{Z}).$$

In particular, since $s = ax_0 + by_0$,

$$g \mid s.$$

Because $s > 0$, this implies

$$\boxed{g \leq s.}$$

Final Result

We have

$$s \leq \gcd(a, b) \quad \text{and} \quad \gcd(a, b) \leq s.$$

Therefore,

$$\boxed{s = \gcd(a, b).}$$

So $\gcd(a, b)$ is exactly the smallest positive integer linear combination of a and b .

6 Important Corollaries

Corollary 31.3: Common divisor divides the GCD

For any integers a and b , if

$$d \mid a \quad \text{and} \quad d \mid b,$$

then

$$\boxed{d \mid \gcd(a, b)}.$$

Proof. By Bézout's characterization, there exist integers x and y such that

$$\gcd(a, b) = ax + by.$$

Since $d \mid a$ and $d \mid b$, the linear combination rule gives

$$d \mid (ax + by).$$

Therefore,

$$\boxed{d \mid \gcd(a, b)}.$$

Corollary 31.4: Scaling rule

For all integers a and b , and every nonnegative integer n ,

$$\boxed{\gcd(an, bn) = n \gcd(a, b)}.$$

Proof. If $n = 0$, then

$$\gcd(an, bn) = \gcd(0, 0) = 0$$

and

$$n \gcd(a, b) = 0 \cdot \gcd(a, b) = 0.$$

So the identity holds.

Now assume $n > 0$. By Bézout's characterization, $\gcd(an, bn)$ is the smallest positive integer of the form

$$anx + bny, \quad x, y \in \mathbb{Z}.$$

Factor out n :

$$anx + bny = n(ax + by).$$

Thus the positive linear combinations of an and bn are exactly n times the positive linear combinations of a and b . Therefore the smallest positive one is

$$n \gcd(a, b).$$

Hence

$$\boxed{\gcd(an, bn) = n \gcd(a, b)}.$$

Corollary 31.5: Cancellation rule

For all positive integers n, a, b , if

$$n \mid ab \quad \text{and} \quad \gcd(a, n) = 1,$$

then

$$\boxed{n \mid b}.$$

Proof. Since $\gcd(a, n) = 1$, Bézout's characterization gives integers x and y such that

$$ax + ny = 1.$$

Multiply both sides by b :

$$abx + nby = b.$$

Now use the assumptions:

$$n \mid ab \implies n \mid abx,$$

and

$$n \mid nby.$$

Here is why those two statements are valid:

$$n \mid ab \implies ab = nt \text{ for some } t \in \mathbb{Z}.$$

Multiplying by x ,

$$abx = ntx = n(tx),$$

so

$$n \mid abx.$$

Also,

$$nby = n(by),$$

so

$$n \mid nby.$$

Therefore n divides their sum:

$$n \mid (abx + nby).$$

But $abx + nby = b$, so

$$\boxed{n \mid b.}$$

7 Worked Example

Problem

Find $\gcd(24, 30)$, then show that every common divisor of 24 and 30 divides it.

Solution

List positive divisors:

$$\text{Div}(24) = \{1, 2, 3, 4, 6, 8, 12, 24\},$$

$$\text{Div}(30) = \{1, 2, 3, 5, 6, 10, 15, 30\}.$$

Common divisors:

$$\{1, 2, 3, 6\}.$$

Therefore,

$$\boxed{\gcd(24, 30) = 6.}$$

Each common divisor 1, 2, 3, 6 divides 6, matching the corollary.

8 Final Summary

Keep These

d common divisor of $a, b \iff d \mid a$ and $d \mid b$.

$d \mid a, d \mid b \implies d \mid (ax + by)$.

$\gcd(a, b)$ = largest positive common divisor of a, b .

$\gcd(a, b) = \min\{ax + by > 0 : x, y \in \mathbb{Z}\}$.

$d \mid a, d \mid b \implies d \mid \gcd(a, b)$.